CHAPTER

EIGHT

Modern Electronics

Lesson 1
- The Semiconductor Crystal.
- The pn Junction.

Lesson 2
- The Transistor.
- Digital and Analog Electronics.

- Questions on each lesson from the Previous Years Exams.
- Test on Chapters 7 and 8

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Chapter 8

Questions on Lesson One

- The Semiconductor Crystal.
- The pn Junction.

Interactive test



The questions signed by ✳ are answered in detail. ● Understand ○ Apply Higher Order Thinking Skills

Important guidelines

Law of mass action: $np = n_i^2$

(Where: (n_i) is the concentration of free electrons or holes in the pure (intrinsic) semiconductor crystal).

Semiconductor

Intrinsic crystals — $n = p = n_i$

Extrinsic crystals (doped)

	n-type	p-type
Impurity element valency	Pentavalent (doners)	Trivalent (acceptors)
Free electrons concentration	$n = N_D^+ + P$ (Electrically neutral) $n \approx N_D^+$	$n = \frac{n_i^2}{N_A^-}$
Holes concentration	$p = \frac{n_i^2}{N_D^+}$	$p = N_A^- + n$ (Electrically neutral) $p = N_A^-$
The relation between the two types of charge carriers	$n > p$	$p > n$
Crystal returns back to its initial state (n = p) when	$N_A^- = N_D^+$	$N_A^- = N_D^+$

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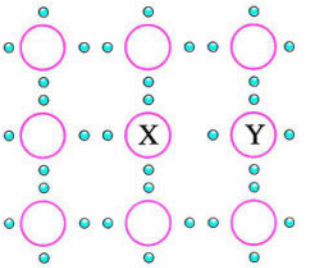
First Multiple choice questions

The semiconductor crystal

1. Pure silicon or germanium crystals become perfectly insulating to electricity at a temperature of ..... .

(a) 0°C (b) 273°C (c) – 273°C (d) 273 K

2. The opposite figure shows a section of pure silicon crystal at room temperature in which the covalent bond between atom (X) and atom (Y) in the crystal is incomplete and that's because one of the electrons of the bond ..... .



(a) has repelled the other electron in the bond

(b) has gained enough energy from the surrounding environment to get free

(c) has moved into an inner energy level in atom (X)

(d) has been captured by atom (Y)

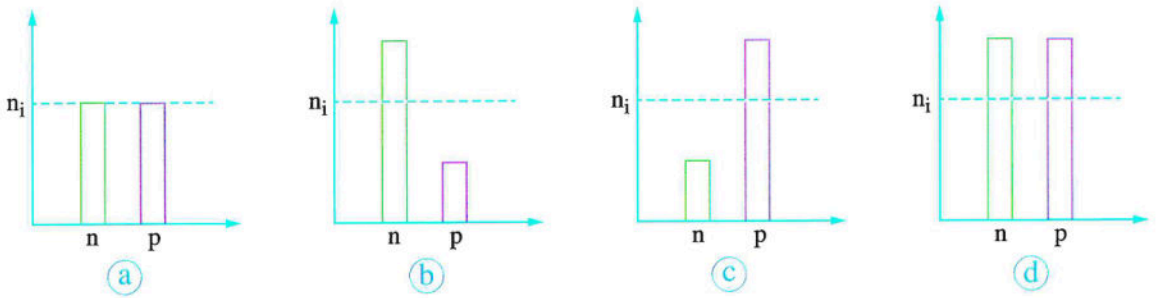
3. If a copper chip and a germanium crystal are cooled down from 300 K to 200 K, the electrical resistance ..... .

(a) increases for each of them (b) decreases for each of them

(c) increases for the copper chip while it decreases for the germanium crystal

(d) decreases for the copper chip while it increases for the germanium crystal

4. In a pure semiconductor crystal at 25°C, the concentration of free electrons = The concentration of holes = n_i, so which of the following graphs represents the concentration of free electrons (n) and that of holes (p) at 50°C?



5. Two intrinsic (pure) silicon crystals A and B are put at temperatures t_1 and t_2 respectively. If $t_2 > t_1$, the mathematical relation that represents the concentration of free electrons (n) and the concentration of holes (p) in both crystals is ..... .

(a) $n_A + n_B = p_A + p_B$ (b) $n_A + p_A = n_B + p_B$

(c) $n_A \times p_A = n_B \times p_B$ (d) $n_B - n_A > p_B - p_A$

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6 Three intrinsic silicon crystals x, y, z where crystal (x) is in a state of thermodynamic equilibrium, crystal (y) temperature is increased, crystal (z) temperature is decreased, which of the three crystals:

(i) Contains positive charge carriers?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

(ii) The charge carriers' concentration in it increases?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

(iii) Its resistivity increases?

(a) Crystal (x). (b) Crystal (y). (c) Crystal (z). (d) The three crystals.

7 The element with which a silicon crystal is doped to form an n-type semiconductor is ..... .

(a) boron (trivalent)
(b) antimony (pentavalent)
(c) nickel (divalent)
(d) aluminum (trivalent)

8 When increasing the percentage of trivalent impurities in a p-type semiconductor crystal, the electrical conductivity of the crystal will ..... .

(a) increase
(b) decrease
(c) doesn't change
(d) be indeterminable

9 A p-type semiconductor crystal in which the concentration of holes equals X, so when adding pentavalent impurities to the crystal by a concentration of $\frac{1}{2}$X, its electric conductivity ..... .

(a) decreases
(b) remains unchanged
(c) increases
(d) can't be determined

10 The dominant (majority) charge carriers in p-type crystal are .....

(a) free electrons (b) holes (c) positive ions (d) negative ions

11 A silicon crystal is doped with atoms of a pentavalent element, so at equilibrium the ratio between the concentration of holes to that of free electrons is ..... .

(a) less than one
(b) greater than one
(c) equal to one
(d) the information is insufficient for determining the answer

12 The n-type semiconductor crystal is ..... .

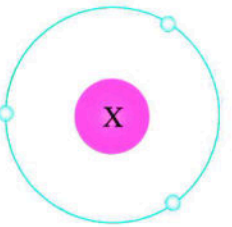
(a) negatively charged
(b) electrically neutral
(c) positively charged
(d) electrically insulator

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13 In an impure semiconductor crystal, if the concentrations of free electrons and holes are n and p respectively, certainly .....

(a) $n > p$ (b) $n < p$ (c) $n = p$ (d) $n \neq p$

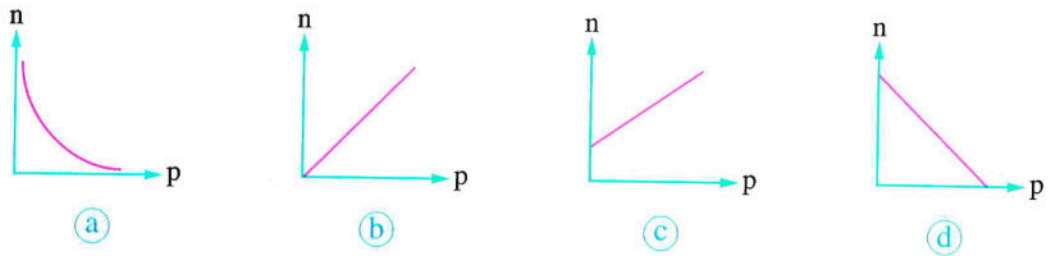
14 The opposite diagram illustrates the electrons of the outermost shell for element $_{49}X$, if a pure semiconductor crystal is doped with atoms of this element, the type and the charge of the resulted crystal respectively will be ..... .



(a) n, neutral (b) p, positive

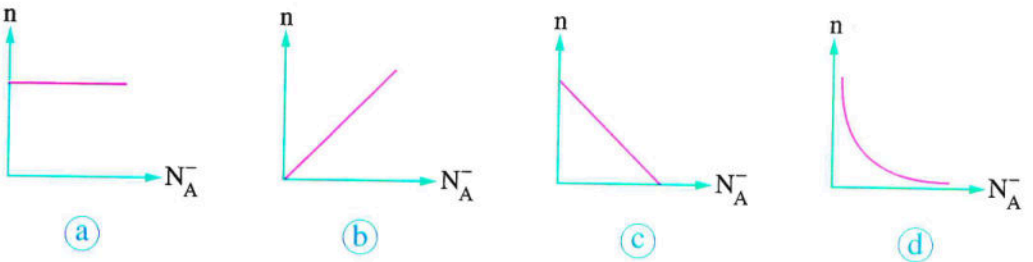
(c) p, neutral (d) n, negative

15 The graph that represents the relation between the concentration of free electrons (n) and the concentration of holes (p) in a pure silicon crystal when increasing its temperature is ..... .



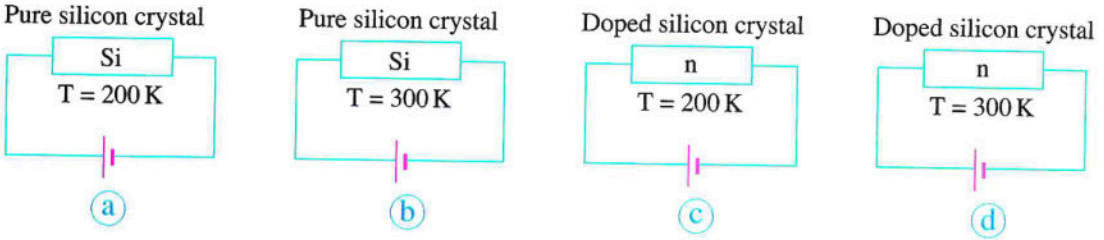
(a) (b) (c) (d)

16 Which of the following graphs represents the relation between the concentration of free electrons (n) and that of aluminum ions (N_A^-) in a p-type semiconductor crystal?



(a) (b) (c) (d)

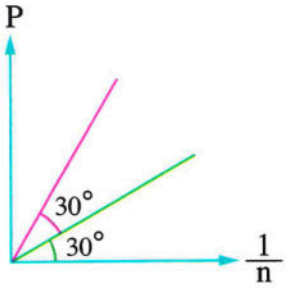
17 Which of the following circuits has the greatest electrical resistance to the flow of electric current?



(a) (b) (c) (d)

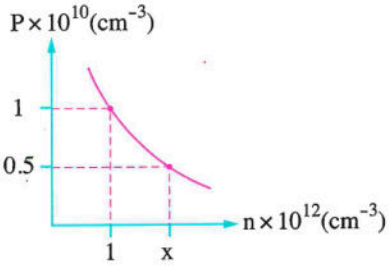
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18 Two intrinsic silicon crystals a, b at different temperatures where $(t_a > t_b)$ were doped with phosphorus atoms, the opposite graph represents the relation between the holes concentration (P) in the doped crystal and the inverse of the free electron concentration $\left(\frac{1}{n}\right)$ in it, then the ratio of the holes concentrations in the two pure crystals before doping $\left(\frac{P_a}{P_b}\right)$ is ..... .



(a) 3 (b) $\sqrt{3}$ (c) $\frac{1}{3}$ (d) $\frac{1}{\sqrt{3}}$

19 A pure semiconductor crystal is doped with phosphorus atoms, and the opposite graph represents the relation between the holes concentration (P) and the concentration of free electrons (n) in the crystal, then:



(i) The concentration of free electrons in the intrinsic semiconductor crystal is equal to ..... .

(a) 10^{10} cm^{-3} (b) 10^{11} cm^{-3}
(c) 10^{12} cm^{-3} (d) 10^{22} cm^{-3}

(ii) The value of (x) on the graph is equal to ..... .

(a) 1.5×10^{12} cm^{-3} (b) 2×10^{12} cm^{-3}
(c) 2.5×10^{12} cm^{-3} (d) 3×10^{12} cm^{-3}

20 Three identical intrinsic silicon crystals (a), (b), and (c) , crystal (b) was doped with aluminum atoms at a concentration of (X) cm^{-3} and crystal (c) was doped with phosphorus atoms at the same concentration of (X) cm^{-3}.
Assuming all of the three crystals are at the same temperature, the correct order of the crystals based on the concentration of free electrons in each is ..... .

(a) $a = b = c$ (b) $a < b < c$ (c) $c = b > a$ (d) $b < a < c$

21 If the concentrations of free electrons and holes in a silicon crystal doped with arsenic are 10^{13} cm^{-3} and 10^{9} cm^{-3} respectively, the concentration of each of free electrons and holes in a pure silicon crystal equals ..... .

(a) 10^{9} cm^{-3} (b) 10^{10} cm^{-3} (c) 10^{11} cm^{-3} (d) 10^{13} cm^{-3}

22 In a semiconductor crystal, the concentration of free electrons is 5×10^{13} cm^{-3} and the concentration of holes is 8×10^{12} cm^{-3}, so this semiconductor crystal is ..... .

(a) n-type (b) p-type (c) positively charged (d) pure

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23 The concentration of either free electrons or holes in each of two identical intrinsic (pure) silicon crystals x and y is 10^{12} cm^{-3}, so if crystal x gets doped with antimony atoms in a concentration of 10^{14} cm^{-3} while crystal y gets doped with aluminum atoms in a concentration of 10^{14} cm^{-3}, the ratio between the concentrations of holes in the two crystals $\left(\frac{p_x}{p_y}\right)$ will be ..... .

(a) 10^{-2} (b) 10^{-4}
(c) 10^{-6} (d) 10^{-8}

24 * The concentration of free electrons or that of holes in a pure silicon crystal is 10^{8} cm^{-3}, if the crystal is doped with aluminum atoms with a concentration of 10^{10} cm^{-3}, hence the concentration of holes and that of free electrons in this case at the complete ionization of the atoms of impurities respectively will be ..... .

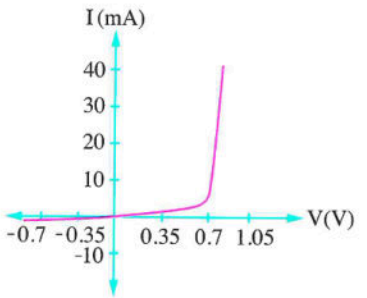
(a) 10^{10} cm^{-3}, 10^{8} cm^{-3} (b) 10^{8} cm^{-3}, 10^{10} cm^{-3}
(c) 10^{6} cm^{-3}, 10^{10} cm^{-3} (d) 10^{10} cm^{-3}, 10^{6} cm^{-3}

The pn junction

25 When connecting a diode in reverse direction to a battery, ...

(a) the potential barrier increases and the resistance increases
(b) the potential barrier decreases and the resistance decreases
(c) the potential barrier increases and the resistance decreases
(d) neither the potential barrier nor the resistance changes

26 The opposite graph represents the relation between the intensity of the current (I) through a semiconductor diode and the potential difference (V) across its terminals, so the potential barrier for this diode is ..... .



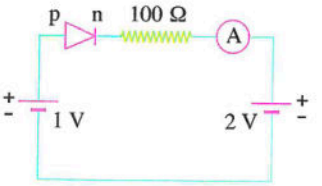
(a) 1.2 V (b) 0.8 V
(c) 0.7 V (d) zero

27 The depletion region of the pn junction has a high electrical resistance because of ..... .

(a) not containing charge carriers
(b) containing a large number of charge carriers
(c) containing only free electrons
(d) containing only holes

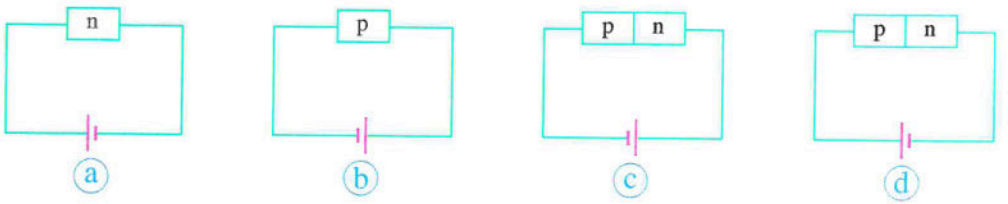
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28 The ammeter reading in the opposite circuit is ..... .



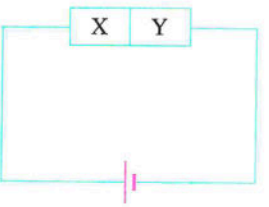
(a) zero
(b) 1 mA
(c) 10 mA
(d) 30 mA

29 Which of the following electric circuits **doesn't** carry electric current?



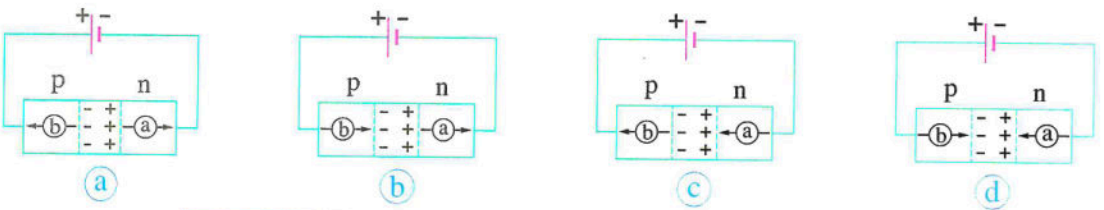
(a) (b) (c) (d)

30 The opposite figure represents a diode made of silicon in which region X contains arsenic atoms and region Y contains boron atoms, hence ..... .



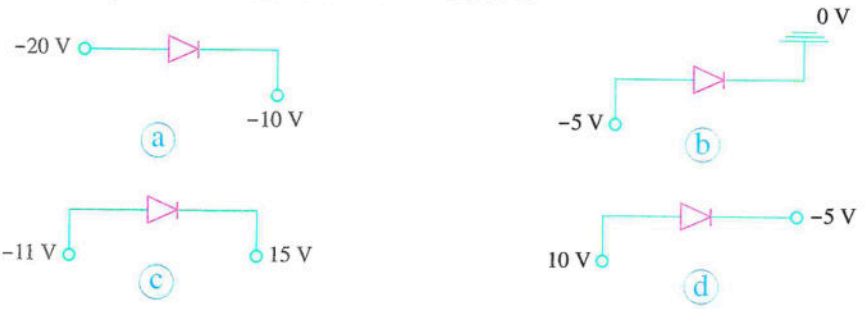
	Type of region X	Type of region Y	Bias
(a)	p	n	forward
(b)	n	p	forward
(c)	p	n	reverse
(d)	n	p	reverse

31 If a and b are referring to the dominant charge carriers in the two regions (n, p) of a diode respectively, which of the following figures represents the directions of their motions in the case of forward bias connection?



(a) (b) (c) (d)

32 The figure that represents a forward biased diode is ..... .



(a) (b) (c) (d)

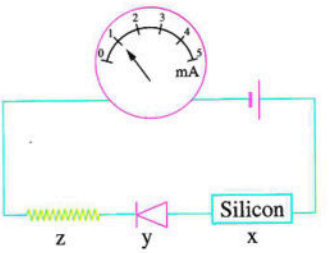
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33 The opposite figure represents a part of an electric circuit, if the resistance of the pn junction is negligible in forward bias and infinite in reverse bias, the intensity of the passing current is ..... .



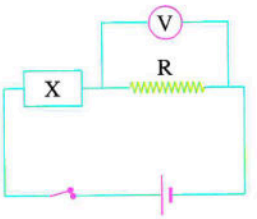
(a) 0 (b) 10^{-2} A (c) 10^{2} A (d) 10^{-3} A

34 In the opposite circuit, which of the following actions leads to an increase in the reading of the milliammeter?



(a) Reversing the polarity of the battery
(b) Reversing the direction of component y
(c) Increasing the temperature of component x
(d) Increasing the temperature of component z

35 In the opposite electric circuit, if the reading of the voltmeter is approximately equal to zero, then component X is ..... .





(a) (b) (c) (d)

36 When a certain electrical component was connected between the terminals of an ohmmeter, the pointer position was as in figure (1) and when reversing the connection of the component between the ohmmeter terminals, the pointer position became as in figure (2), then this component is ..... .

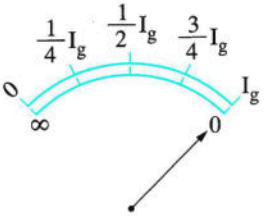


Figure (1)

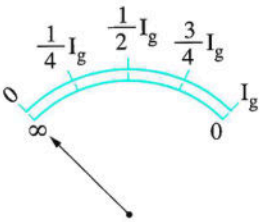
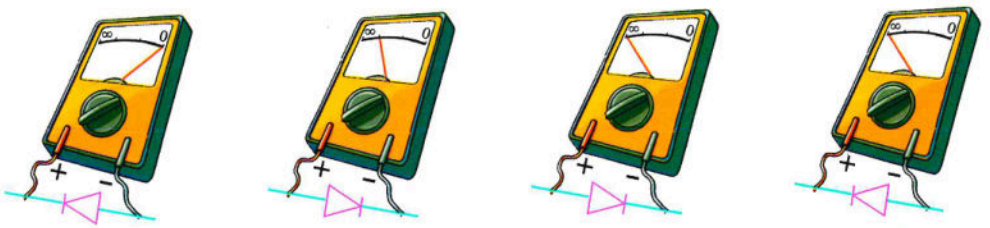


Figure (2)

(a) a pure silicon crystal (b) an n-type silicon crystal
(c) a p-type silicon crystal (d) a diode

37 An ohmmeter gives zero reading when its terminals are connected together, so if a pn junction has negligible resistance in forward bias and infinite resistance in reverse bias, which of the following figures illustrates the correct deflection of the ohmmeter pointer when the pn junction is connected between its terminals?



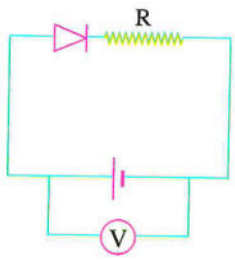
(a) (b) (c) (d)

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38 * A diode with a forward bias resistance of 100 Ω and a reverse bias resistance of infinity is connected to a source of potential difference 5 V such that the p-side of the diode gets connected with the positive potential at a time then reversed so that the p-region gets connected to the negative potential at another time, so the current intensities through the diode in the two cases respectively are ..... .

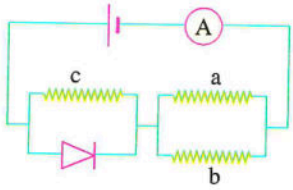
(a) 0, 0 (b) 0.05 A, 0 (c) 0, 0.05 A (d) 0.05 A, 0.05 A

39 * In the opposite electric circuit, the voltmeter reading is 1.5 V, so if the potential difference across the diode terminals is 0.5 V and the electric power consumed through it equals 100 mW, so the resistance of R is ..... .



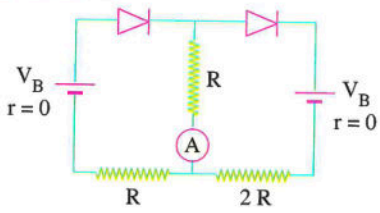
(a) 2.5 Ω (b) 5 Ω
(c) 7.5 Ω (d) 10 Ω

40 * The electric circuit in the opposite figure consists of a dry cell of emf V_B whose internal resistance is negligible, three identical ohmic resistances (a, b, c) and a diode with a forward bias resistance equal to the value of one of the three ohmic resistances and an infinite resistance in reverse bias, hence the ratio between the readings of the ammeter before and after reversing the diode connection respectively equals ..... .



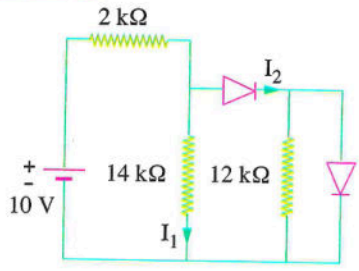
(a) $\frac{1}{2}$ (b) $\frac{2}{1}$ (c) $\frac{2}{3}$ (d) $\frac{3}{2}$

41 In the shown electric circuit, if the resistance of the diode is negligible in forward bias and infinite in reverse bias, the reading of the ammeter is ..... .



(a) $\frac{V_B}{3R}$ (b) $\frac{V_B}{2R}$
(c) $\frac{2V_B}{3R}$ (d) $\frac{3V_B}{2R}$

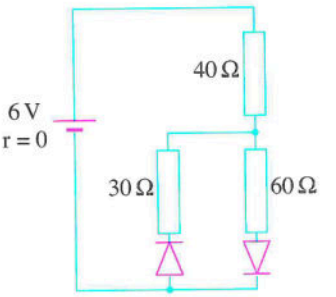
42 In the opposite circuit, if the resistance of the pn junction can be neglected in forward bias, while considered infinite in reverse bias, so the values of I_1 and I_2 respectively are ..... .



(a) 0, 0 (b) 5 mA, 5 mA
(c) 5 mA, 0 (d) 0, 5 mA

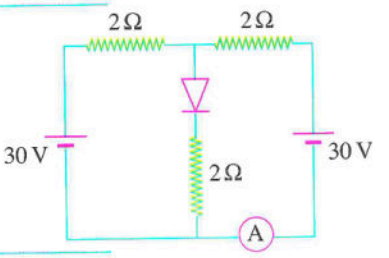
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43 * In the opposite diagram, considering that the internal resistance of the electric source is negligible and supposing that the resistances of the diodes in forward connection are negligible while in reverse connection are infinite, the intensity of the current in the 40 Ω resistor is ..... .



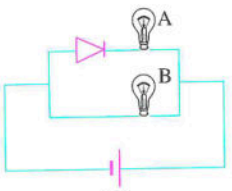
(a) 0 (b) 0.06 A
(c) 0.1 A (d) 0.12 A

44 What happens to the reading of the ammeter (A) when reversing the connection of the diode in the circuit shown in the opposite figure?

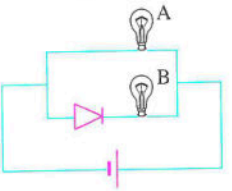


(a) Increases. (b) Decreases.
(c) Becomes zero. (d) Doesn't change.

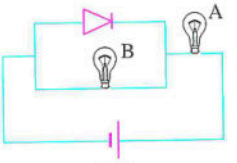
45 Two identical lightbulbs (A, B) are connected with a pn junction to a battery in several ways, in which of the following figures does bulb (A) light up?



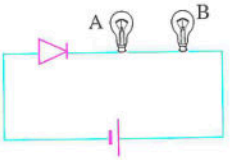
(1)



(2)



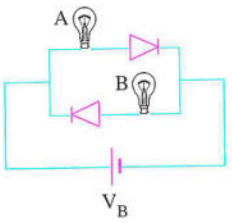
(3)



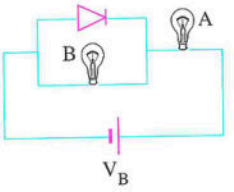
(4)

(a) (1), (3) (b) (2), (3) (c) (2), (4) (d) (3), (4)

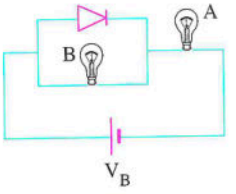
46 In the following electric circuits, given that the two lightbulbs have the same resistance which is five times as large as the resistance of the pn junction in forward bias, so in which circuit is the brightness of bulb A less than that of bulb B?



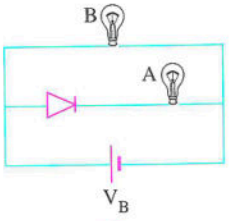
(a)



(b)

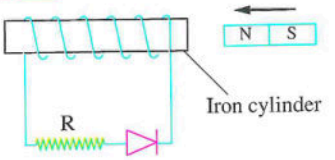


(c)



(d)

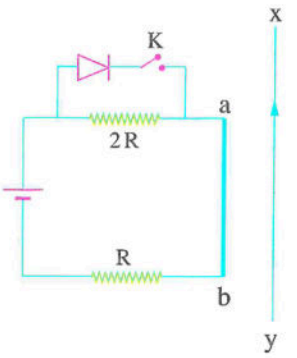
47 A coil is wound around an iron cylinder and connected to an ohmic resistance (R) and a diode, as shown in the opposite figure. When a magnet approaches the coil, ...



	In the coil	In the resistance
(a)	An emf is induced	An electric current flows
(b)	An emf is induced	No electric current flows
(c)	No emf is induced	An electric current flows
(d)	No emf is induced	No electric current flows

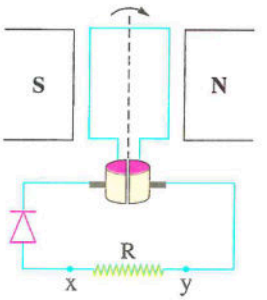
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48 A long straight wire (xy) carries an electric current and is placed parallel to another straight wire (ab) with resistance (R) connected to an electric circuit as shown in the opposite figure, if the diode in the circuit is assumed to have negligible resistance in forward bias and infinite in reverse bias, what happens to the magnitude of the mutual magnetic force between the two wires (xy) and (ab) when the switch (K) is closed?



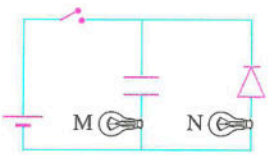
(a) It decreases to half. (b) It doubles.
(c) It vanishes. (d) It doesn't change.

49 The opposite figure shows a dynamo connected to an ohmic resistance (R) and a diode, then the resistance (R) ..... .



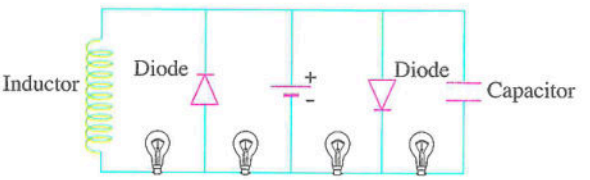
(a) has an electric current always passing through it from point x to y
(b) has an electric current always passing through it from point y to x
(c) has an alternating current passing through it
(d) doesn't have electric current

50 Two identical lamps (N, M) are connected to a battery, a capacitor and a pn junction as shown in the opposite electric circuit, which of the two lamps flashes at the moment of closing the switch?



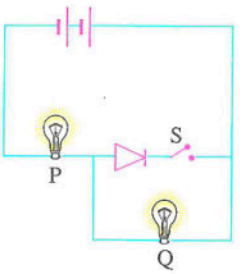
(a) Only M. (b) Only N.
(c) M and N. (d) No one will glow.

51 The opposite electric circuit contains multiple electric components and multiple identical lightbulbs, so the number of lightbulbs that are glowing in the circuit is ..... .



(a) 1 (b) 2 (c) 3 (d) 4

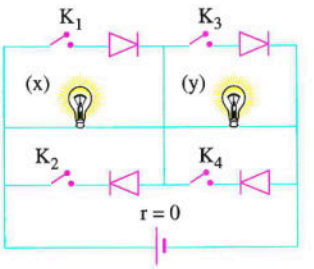
52 Two identical lamps P, Q are connected in an electric circuit with a pn junction as shown in the opposite figure, what happens to the lights of the two lamps when closing switch S?



(a) Both go off. (b) P stays on, Q goes off.
(c) P goes off, Q stays on. (d) Both stay on.

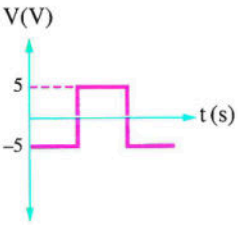
CamScanner

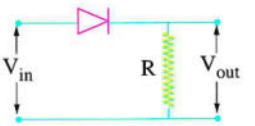
53 Two identical light bulbs x and y, are connected with several diodes through four switches as shown in the opposite electrical circuit. Assuming that the resistance of each diode is R when forward-biased and infinite when reverse-biased, then which two switches, when closed, will increase the brightness of bulb (x) and decrease the brightness of bulb (y)?

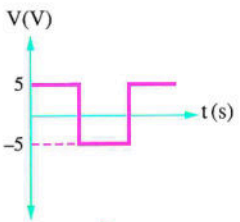


(a) Switches (K_1) and (K_3).
(b) Switches (K_2) and (K_4).
(c) Switches (K_1) and (K_4).
(d) Switches (K_2) and (K_3).

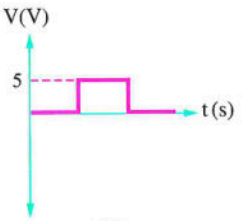
54 In the opposite figure, an electric signal passes through an ideal diode, so which of the following graphs represents the output signal (V_{out}) across resistor R?



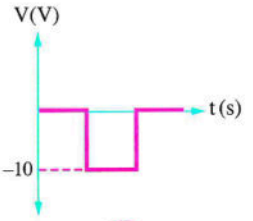




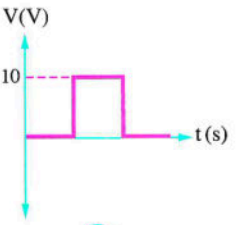
(a)



(b)

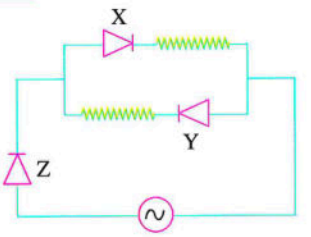


(c)



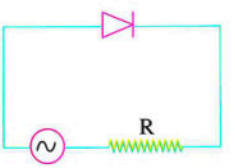
(d)

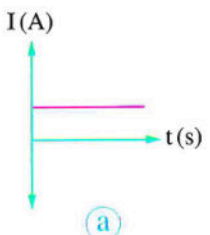
55 Some types of diodes are designed to emit light when connected in forward bias only. These diodes are called LEDs (light-emitting diodes). If three of these LEDs are connected to a low frequency AC source as shown in the opposite circuit, which of the following choices is correct?



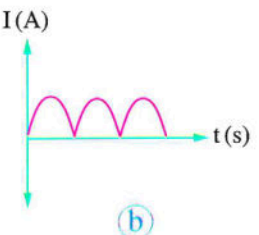
(a) LED X lights up only when LED Z lights up.
(b) LED Z lights up only when LED X goes off.
(c) LED Y lights up only when LED X goes off.
(d) The three LEDs will always light up.

56 From the opposite circuit, the graph that represents the relation between the electric current intensity (I) through resistor R and time (t) is ..... .

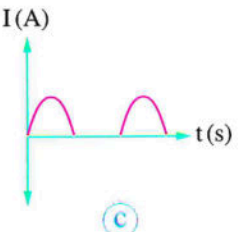




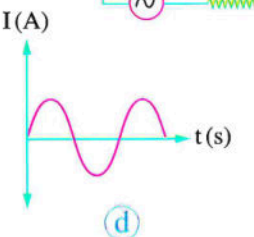
(a)



(b)



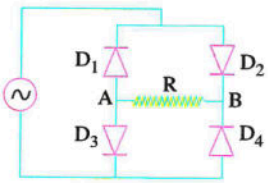
(c)

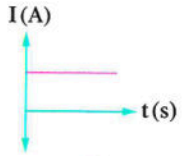


(d)

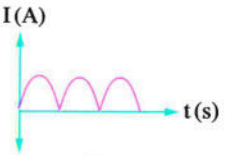
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57 The opposite figure shows four diodes connected in an electric circuit, hence the graph that represents the variation of the current intensity (I) through resistor R versus time (t) is ..... .

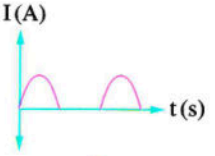




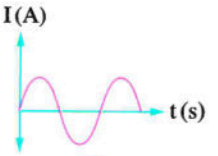
a



b

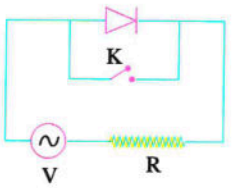


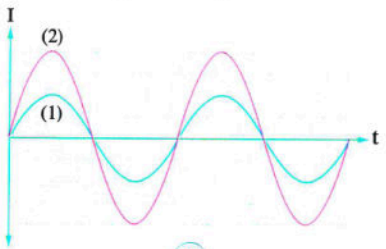
c



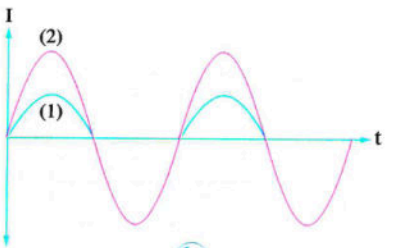
d

58 In the opposite circuit diagram, given that the resistance of the diode in forward bias equals R and in reverse bias is infinite, so the graph in which curve (1) and curve (2) represent the change of current intensity (I) versus time (t) through resistor R before and after closing switch K respectively is ..... .

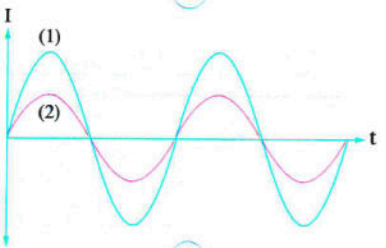




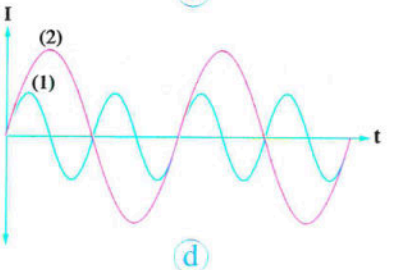
a



b

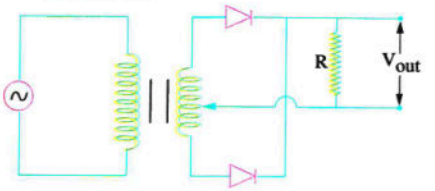


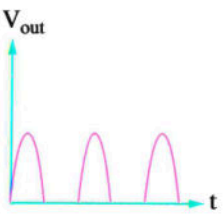
c



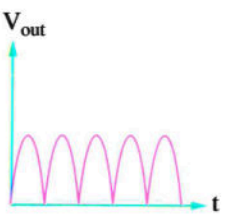
d

59 The graph that expresses the relation between the output voltage (V_{out}) and the time (t) for the opposite electric circuit is ..... .

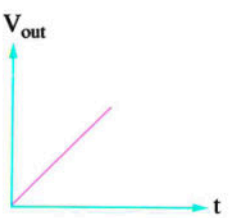




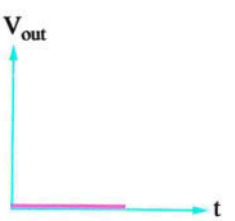
a



b



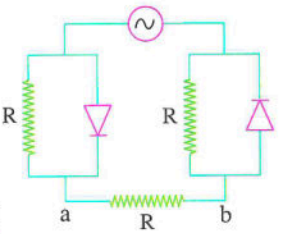
c

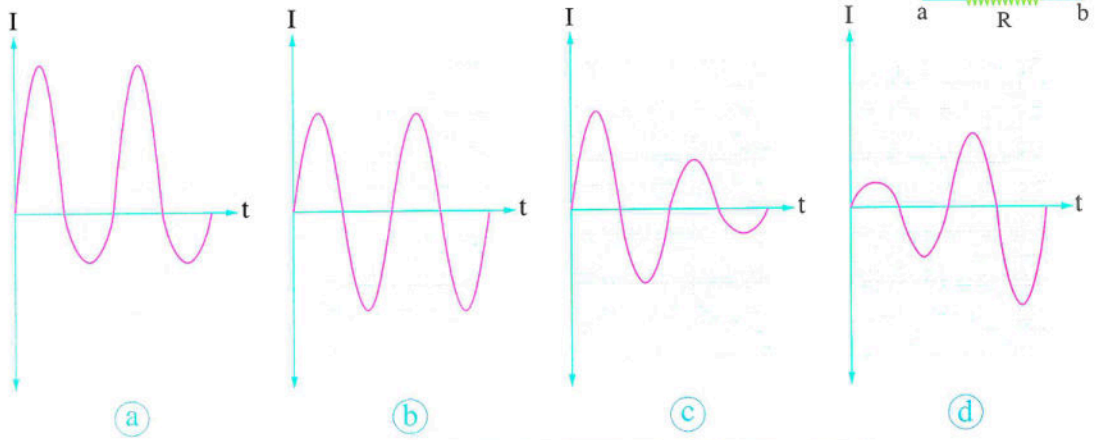


d

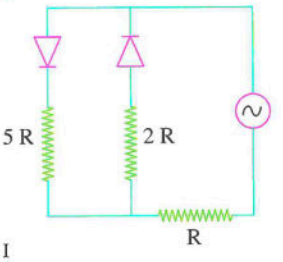
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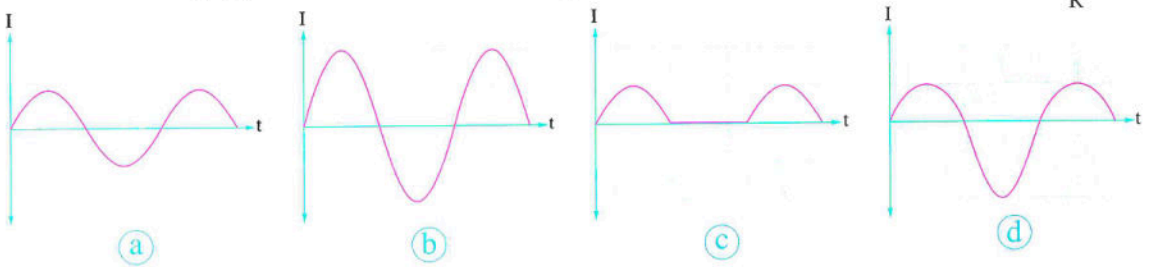
60 In the opposite circuit, if the resistances of the diodes are negligible in forward bias and infinite in reverse bias, the graph which represents the electric current through the resistance between the two points a and b is ..... .





61 In the opposite figure; if the diodes have negligible resistances in forward bias and infinite resistance in reverse bias, which of the following graphs represents the change of the current intensity (I) in resistor R with time (t)?





Second Essay questions

Semiconductor crystal

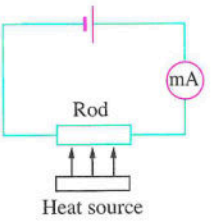
1 **Why** is pure silicon a poor conductor of electric current at low temperatures?

2 **Give reasons for:** It is not preferable to heat the pure semiconductor for a high temperature to increase its conductivity to electric current.

3 **Using the opposite figure, explain what happens to the reading of the milliammeter in the two following cases:**

(a) If the rod is made of copper.

(b) If the rod is made of silicon.



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4 **Explain why**, when a pure silicon crystal is doped with phosphorus atoms, the electrical conductivity of the crystal increases even though its total electric charge does not change.

5 **In a semiconductor crystal of p-type:**

(a) How many valence electrons should be in an atom of the impurity substance?

(b) Explain, is doping the pure crystal with atoms of impurity substance makes it positively charged.

(c) What is the type of the dominant charge carriers in the p-type semiconductor?

6 If the concentration of free electrons or that of holes in a pure silicon crystal is 10^{10} cm^{-3}, then phosphorus atoms are added to the crystal with a concentration of 10^{12} cm^{-3}, **so calculate:**

(a) The concentrations of free electrons and that of holes in this case.

(b) The concentration of aluminum atoms that are required to be added to the silicon crystal to return to be as if it is a pure crystal.

7 Three crystals of silicon (x, y, z) are of the same size at the same temperature, crystal (x) is pure, crystal (y) is doped with impurities of phosphorus atoms with a concentration of 2×10^{10} cm^{-3} and crystal (z) is doped with impurities of boron atoms with a concentration of 10^{10} cm^{-3}, **explain why** the conductivity of crystal (z) is greater than that of crystal (x) and less than that of the crystal (y).

8 A silicon crystal is doped with phosphorus atoms at a concentration of 2×10^{10} cm^{-3}, **explain what happens to its conductivity when adding:**

(a) Phosphorus atoms with a concentration of 10^{10} cm^{-3}.

(b) Aluminum atoms with a concentration of 10^{10} cm^{-3}.

The pn junction

9 **You have a pn junction and a battery:**

(a) Explain how they can be connected so that the electric field of the battery works:

(1) In the same direction as the electric field inside the junction.

(2) In a direction opposite to the electric field inside the junction.

(b) Compare the results of the method of connection in each case on:

(1) Potential barrier.

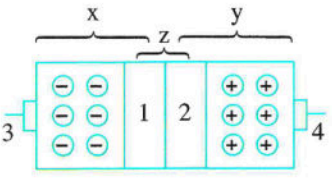
(2) The thickness of the depletion region.

(3) The flow of current through the junction.

10 The opposite diagram represents a pn junction:

(a) What is the type of the semiconductor crystal denoted by (x) and that denoted by (y)?

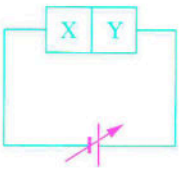
(b) What is the name of region (z)?



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(c) In the case of forward connection, **which** pole of the battery should be connected to terminal (3)?

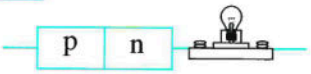
11 The opposite circuit contains a diode in which region X is doped with antimony and region Y is doped with boron, and then the junction is connected to a battery as shown in the figure:



(a) **Is** the junction forward biased or reverse biased?

(b) **Plot the graph** that represents the relation between the current intensity (I) and the potential difference (V) across the junction.

12 The opposite figure represents a diode connected in series to a small lamp:



(a) **Complete the diagram** of the electric circuit by connecting a DC source to light up the lamp **with explanation**.

(b) **What will happen if** the connection of the circuit terminals is reversed with respect to the DC source?

(c) **Determine with explanation** the type of the current that passes through the lamp, if the DC source is replaced by an alternating current source.

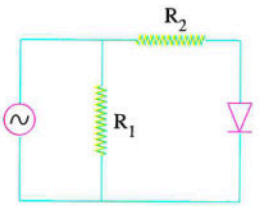
13 **Compare between each of the following:**

(a) Diffusion current and drift current in the diode (in terms of: the direction of free electrons - the cause of occurrence of each of them).

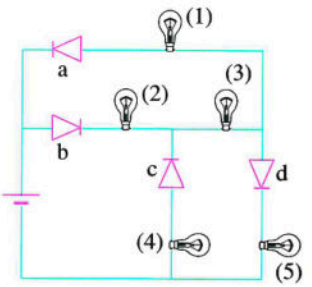
(b) The pn junction and the ohmic electric resistance (in terms of: charge carriers - the direction of electric current - the effect of heating).

14 **Show how** the polarity of pn junction can be determined.

15 In the opposite electric circuit, **why** are the types of electric currents passing in resistors R_1 and R_2 different?



16 In the opposite circuit, a number of identical pn junctions and identical lamps are connected to an electric cell:



(a) **Which** lamp lights up? **And why**?

(b) **Which** of the four pn junctions if its connection method is reversed, no lamp lights up? **Explain your answer.**

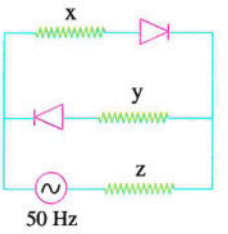
(c) **What lamps are lit with equal brightness when reversing the method of connecting:**

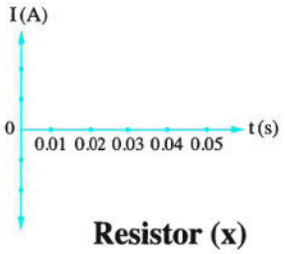
(i) The electric cell? **(ii)** Both pn junctions (a) and (c)?

Explain your answer.

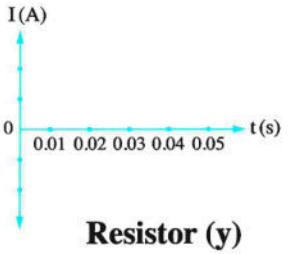
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17 Three identical resistors (x, y, z) and two identical diodes are connected to an alternating electrical source of frequency 50 Hz as shown in the opposite figure, **use the following graphs to graphically represent the current passing through each of the three resistors during 0.05 s.**

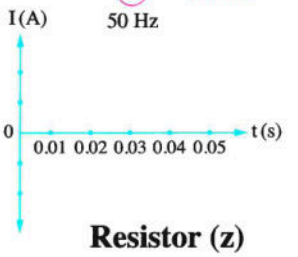




Resistor (x)

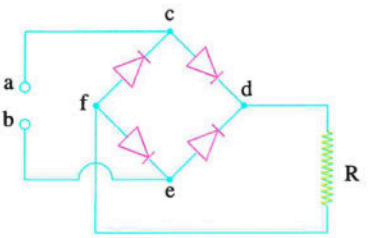


Resistor (y)



Resistor (z)

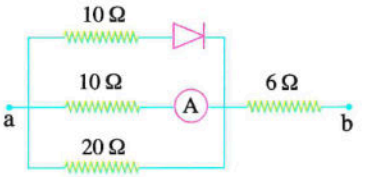
18 The opposite figure represents an electronic circuit containing four pn junctions, ohmic resistor R, and an electric source (ab):



(a) **Trace** with letters the path of electric current through the circuit when the electric potential of:
(1) terminal a is positive.
(2) terminal b is positive.

(b) **According to your answer in part (1), could** the current passing in resistor R be used in recharging a car battery? **And why?**

19 The electrical circuit in the opposite figure shows several resistors, an ammeter, and a diode connected together. If a battery of electromotive force 5 volts that has a negligible internal resistance is connected between the two points a, b. Considering that the resistance of the pn junction is negligible in the case of forward connection and infinite in the case of reverse connection, **Calculate the reading of the ammeter when:**



(a) Potential V_b < Potential V_a

(b) Potential V_b > Potential V_a

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Questions of the Previous Years Exams on Lesson 1

The semiconductor crystal

1 When the temperature of a pure germanium crystal (Ge) is lowered to zero degree Celsius (0°C), its electrical conductivity ..... . *(Exp. June 21)*

(a) increases (b) decreases (c) vanishes (d) does not change

2 Given that, the concentration of the free electrons in a pure germanium crystal in the case of dynamic equilibrium equals 2×10^8 cm^{-3}, hence the expected concentration of holes is ..... . *(1st session 21)*

(a) equal to 2×10^8 cm^{-3}
(b) greater than 2×10^8 cm^{-3}
(c) less than 2×10^8 cm^{-3}
(d) equal to zero

3 Assuming that the temperature of both pure silicon crystal (Si) and a copper wire is dropped to the absolute zero (0 K), so the electrical conductivity ..... . *(2nd session 21)*

(a) vanishes for Si but increases for copper
(b) vanishes for both Si and copper
(c) increases for both Si and copper
(d) increases for Si but vanishes for copper

4 The table illustrates four samples of the same pure semiconductor material at different temperatures, which choice is the correct arrangement for the crystal samples temperatures?

Sample	Temperature	Charge carriers concentration
W	T_W	1.6×10^{16} m^{-3}
X	T_X	1.5×10^{11} cm^{-3}
Y	T_Y	1.6×10^{15} m^{-3}
Z	T_Z	1.5×10^{10} cm^{-3}

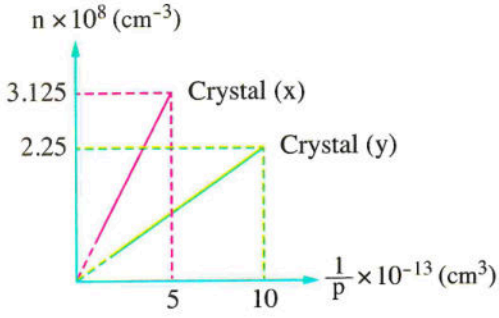
(a) $T_W > T_Y > T_X > T_Z$
(b) $T_X > T_W > T_Z > T_Y$
(c) $T_Z > T_X > T_Y > T_W$
(d) $T_Y > T_Z > T_W > T_X$ *(1st session 22)*

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5 The graph illustrates the relation between the free electrons concentration (n) and the reciprocal of the holes concentration $\left(\frac{1}{p}\right)$ for two extrinsic semiconductor crystals (x) and (y), so the ratio between:

$$\frac{\text{Free electrons concentration in the pure crystal x } [n_{ix}]}{\text{Holes concentration in the pure crystal y } [n_{iy}]}$$

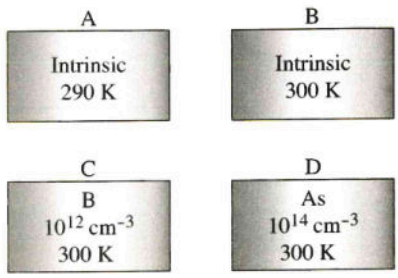
= ..... .



(2nd session 22)

(a) $\frac{25}{9}$ (b) $\frac{25}{36}$ (c) $\frac{5}{6}$ (d) $\frac{5}{3}$

6 The figure represents four identical silicon sheets having equal dimensions (the temperature of each one, the type of impurity and its concentration if exists is recorded), so the arrangement of them according to their electric conductivity from higher to lower is ..... .

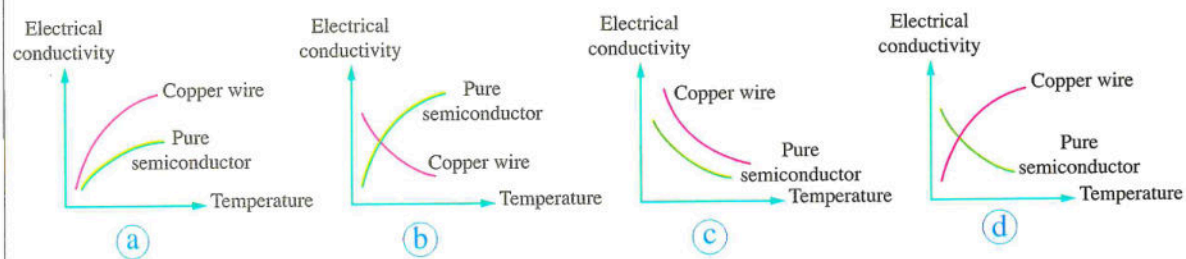


(a) A > B > C > D (b) D > C > B > A

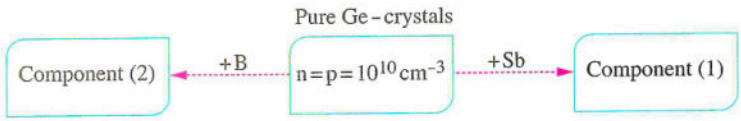
(c) B = C = D > A (d) C = D > B > A *(Exp. 23)*

7 Which graph from the choices illustrates the relation between the electrical conductivity for a pure semiconductor crystal and a copper wire as the absolute temperature changes?

(1st session 23)



8 The following diagram shows the doping processes for pure Ge-crystal to increase the conductivity. *(1st session 24)*



If the concentration of the impurities added in each case is 10^{12} cm^{-3}, so ..... .

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	Component (1)	Component (2)	Ratio $\frac{p_1}{p_2}$	Ratio $\frac{n_1}{n_2}$
(a)	n-type	p-type	10^4	10^{-4}
(b)	n-type	p-type	10^{-4}	10^4
(c)	p-type	n-type	10^4	10^{-4}
(d)	p-type	n-type	10^{-4}	10^4

9 If the concentration of holes in pure semi-conductor crystal was 10^{11} cm^{-3}, then it doped by adding impurity atoms till the concentration of the holes became 10^9 cm^{-3}, which choice is correct? *(2nd session 24)*

	The concentration of electrons in the doped crystal	The impurity atoms
(a)	10^2 cm^{-3}	Phosphorus
(b)	10^2 cm^{-3}	Aluminum
(c)	10^{13} cm^{-3}	Boron
(d)	10^{13} cm^{-3}	Antimony

10 The element that does not produce a p-type semiconductor when doped into a silicon crystal is ..... . *(Practice 25)*

(a) boron (B^{+3})
(b) antimony (Sb^{+5})
(c) nickel (Ni^{+2})
(d) aluminum (Al^{+3})

11 When the temperature of a copper wire and a silicon crystal is raised, the electrical conductivity ..... . *(Practice 25)*

(a) increases for copper and decreases for silicon
(b) decreases for copper and increases for silicon
(c) increases for both
(d) decreases for both

12 The resistivity of intrinsic germanium crystal can be increased by ..... . *(Practice 25)*

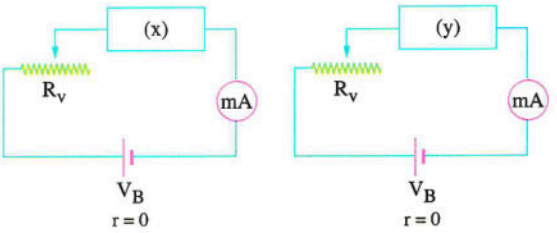
(a) adding a pentavalent impurity
(b) decreasing its temperature
(c) exposing it to a high frequency radiation
(d) adding a trivalent impurity

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13 Two identical intrinsic semiconductor crystals A, B their temperatures are T_1 and T_2 respectively, they are placed in a medium of temperature T_o where $T_1 > T_o > T_2$.
At thermal equilibrium with the medium for each of the crystals, the conductivity of crystal A ..... and the resistivity of crystal B ..... . *(Practice 25)*

(a) decreases, decreases
(b) increases, increases
(c) increases, decreases
(d) decreases, increases

14 The figure shows two different crystals (x) and (y), at a certain temperature each of them is connected separately in a closed electric circuit as shown in the figure, one of them is crystal x for a pure semiconductor, and the other is crystal y for a metallic conductor. On increasing the taken resistance from R_V, the milliammeter (mA) reading decreases. Which of the following actions causes the reading of milliammeter to return to the original case?



(a) Increase the temperature of crystal x and decrease the temperature of crystal y
(b) Increase the temperature of crystal y and decrease the temperature of crystal x
(c) Increase the temperature of both crystals x and y
(d) Decrease the temperature of both crystals x and y *(Practice 26)*

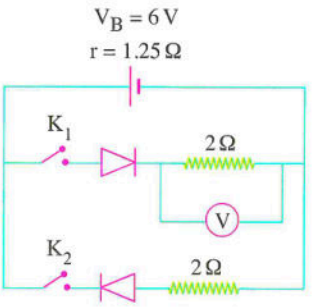
15 In a pure semiconductor crystal, if the concentration of free electrons or holes is 1.5×10^{12} cm^{-3} and pentavalent impurity atoms are added at a concentration of 2×10^{13} cm^{-3} and trivalent impurity atoms are added at a concentration of 1×10^{13} cm^{-3} then ..... . *(Practice 26)*

	Type of the crystal after doping	Concentration of holes after completion of the doping process
(a)	P-type	1×10^{13} cm^{-3}
(b)	P-type	2.25×10^{11} cm^{-3}
(c)	N-type	1×10^{13} cm^{-3}
(d)	N-type	2.25×10^{11} cm^{-3}

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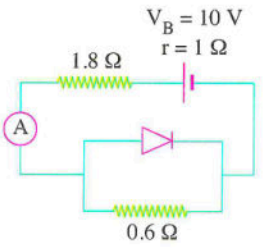
The pn junction

16 In the shown electric circuit. On closing the two keys (K_1), (K_2), the voltmeter reads ..... .
(where the resistance of each diode in forward bias connection = 0.75 Ω and equals (∞) at reverse bias)



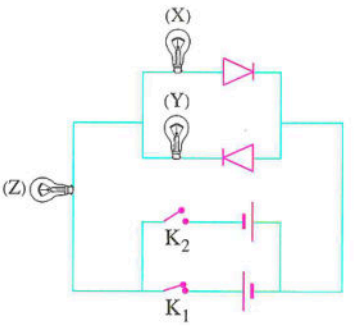
(a) 3 V (b) 0 V
(c) 6 V (d) 4 V *(1st session 22)*

17 In the shown electric circuit, by considering the resistance of the diode in forward bias connection is 0.3 Ω and its resistance in reverse bias connection is infinite (∞), so, the reading of the ammeter equals ..... . *(2nd session 22)*



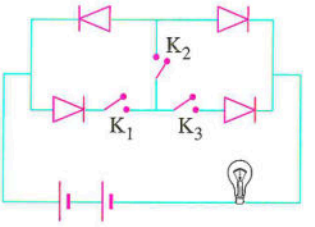
(a) 2.94 A (b) 3.33 A
(c) 2.71 A (d) 3.57 A

18 The figure illustrates an electric circuit that contains three bulbs X, Y and Z connected as in the figure, when K_1 is opened and K_2 is closed. Which of the following choices represents the correct change in the lighting of bulbs? *(Exp. 23)*



(a) Bulb (Y) lights and bulb (X) remains lighted.
(b) Bulb (X) is switched off and bulb (Z) is switched off too.
(c) Bulb (Y) doesn't light and bulb (Z) is switched off.
(d) Bulb (X) is switched off and bulb (Z) remains lighted.

19 According to the given figure:
The resistance of the diode in case of forward connection is 2 Ω and its resistance in backward (reverse) connection is infinite.
Which of the following choices is the correct answer to obtain the largest consumed power in the electric lamp?



	K_1	K_2	K_3
(a)	Closed	Closed	Closed
(b)	Closed	Opened	Opened
(c)	Closed	Closed	Opened
(d)	Closed	Opened	Closed

(1st session 23)

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20 The value of the intensity of the electric field established inside the diode at a certain temperature becomes constant when ..... . *(2nd session 23)*

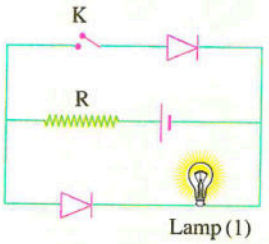
(a) all the holes transfer from the p-region to the n-region

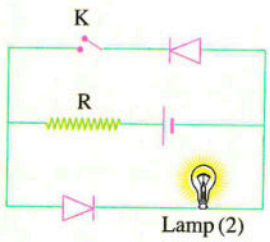
(b) all the free electrons transfer from the n-region to the p-region

(c) the diffussion current equals the drift current

(d) both regions of the diode become electrically neutral

21 If the resistance of the diode in forward bias is neglected and in reverse bias is infinite. *(1st session 24)*

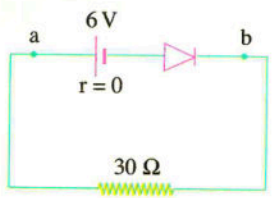




So when the key is closed in the two circuits, ...

	Lamp (1)	Lamp (2)
(a)	goes off	does not change
(b)	glows brighter	goes off
(c)	dims	glows brighter
(d)	does not change	dims

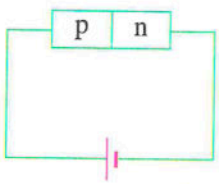
22 The opposite figure shows a battery (of negligible internal resistance) connected in series with a diode and an ohmic resistance, (knowing that the resistance of the diode is negligible in forward connection and infinite in case of backward connection).



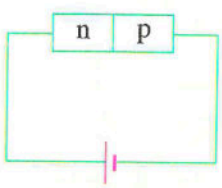
Then the potential difference between the two points a and b is ..... . *(2nd session 24)*

(a) 3 V (b) 0 V (c) 2 V (d) 6 V

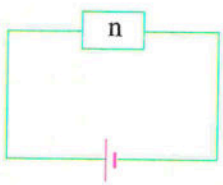
23 Which of the following electrical circuits has the greatest electrical resistance to the flow of electric current? *(Practice 25)*



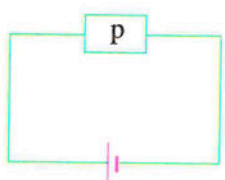
(a)



(b)



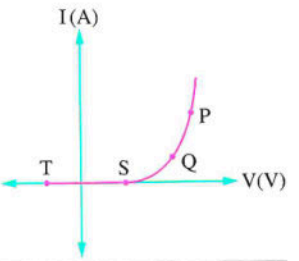
(c)



(d)

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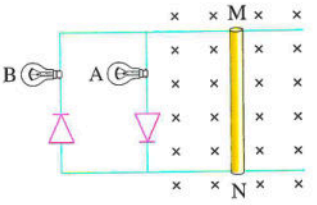
24 The graph illustrates the (I,V) characteristics curve of a diode, so at which of the points shown on the graph is the diode's resistance at its highest value? *(Practice 25)*



(a) S (b) P

(c) Q (d) T

25 In the opposite figure, the electric bulb (A) lights when the wire MN moves ..... . *(Practice 25)*



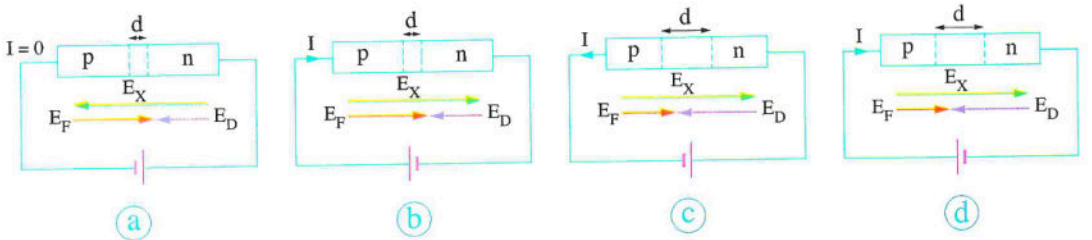
(a) to the right of the page

(b) to the left of the page

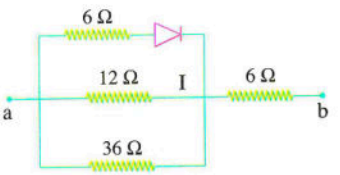
(c) perpendicular into the page

(d) perpendicular out of the page

26 A diode is connected to a DC source, which of the following choices shows correctly the direction of the conventional current (I), thickness of the depletion layer (d), the electric field strength and its directions due to the battery (E_X), the internal electric field (E_D) and the resultant of the produced electric field (E_F) which is acting on the p-n junction? *(Practice 25)*



27 The opposite figure shows a part of an electric circuit, if the electric potential at a is V_a and at b is V_b , where V_b is greater than V_a, so the intensity (I) of the current passing through the resistance 12 Ω in terms of V_a, V_b is determined from the relation ..... . *(Practice 25)*



(where: resistance of the diode is negligible in case of forward connection and infinity in case of reverse connection)

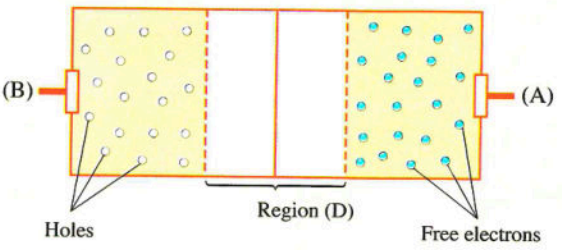
(a) $\frac{5\,(V_b - V_a)}{48}$ (b) $\frac{V_b - V_a}{32}$

(c) $\frac{V_b - V_a}{20}$ (d) $\frac{V_b - V_a}{15}$

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28 The diagram shows a diode (pn junction) not connected to a voltage source.

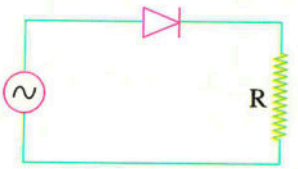
What happens when terminal (A) is connected to the positive pole of an electric cell and terminal (B) is connected to its negative pole? *(1st session 25)*

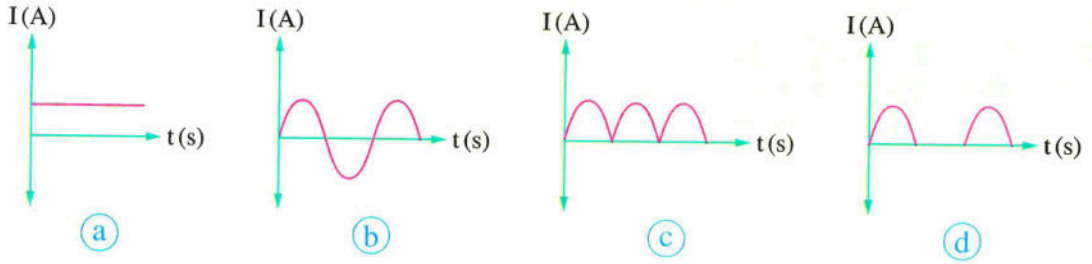


	Region (D)	Free electrons and holes
(a)	Its thickness increases	Move towards region D
(b)	Its thickness decreases	Accumulate near terminals A and B
(c)	Its thickness decreases	Move towards region D
(d)	Its thickness increases	Accumulate near terminals A and B

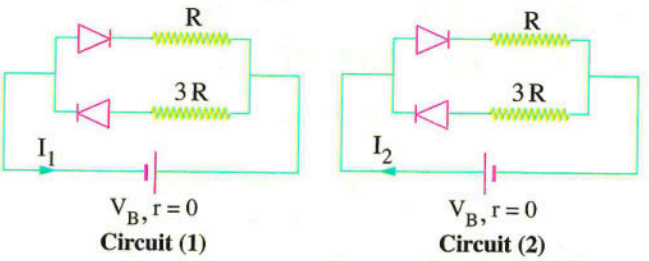
29 In the circuit shown in the figure, a diode is connected to an alternating current source and an ohmic resistor R.

Which of the following graphs represents the relation between the current (I) through the ohmic resistor (R) and the time(t)? *(2nd session 25)*





30 The figure represents two electric circuits, (1) and (2). If the diode's resistance in forward bias is equal to R and infinite in the case of reverse bias, the ratio between the current intensities in the two circuits $\left(\frac{I_2}{I_1}\right)$ = ...



(Practice 26)

(a) $\frac{1}{2}$ (b) $\frac{1}{1}$ (c) $\frac{2}{1}$ (d) $\frac{4}{1}$

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